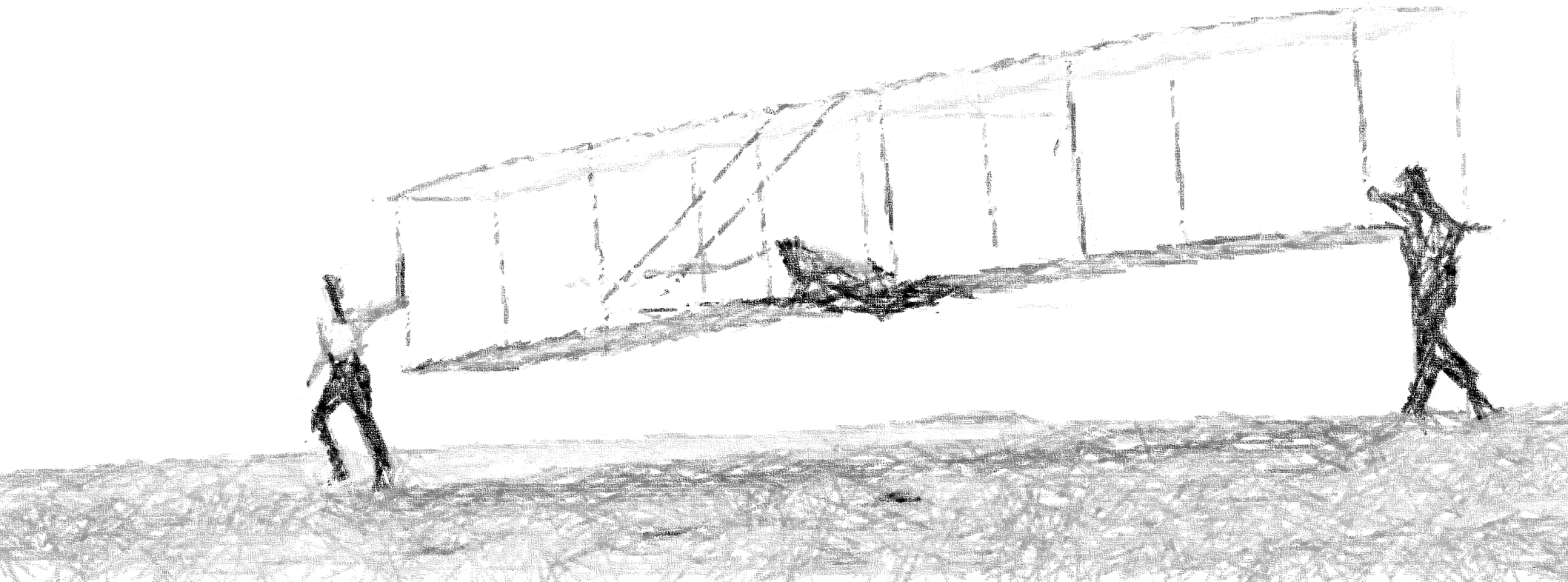




Mechanics applied to Aerospace Engineering

Course 251-14165 – BSc Aerospace Engineering

1. Fundamentals and particle kinematics

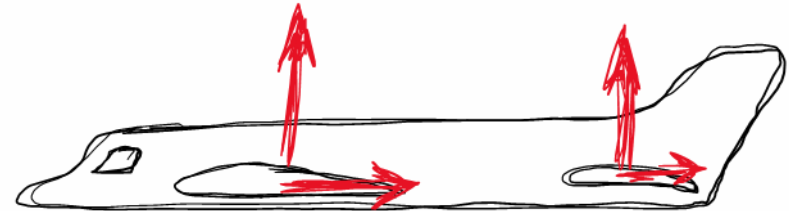


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Mechanical modeling in Aerospace Engineering

- All design and analysis activities in aerospace engineering require developing a *model* of the reality that we want to study



- The model is a simplification of reality that we can work with mathematically. The model is *not* the reality: it retains only the important aspects we are interested in
 - Models can range from very crude to very elaborate
 - We must always validate the conclusions we get from our model with reality
- In this course we will use two types of models to represent reality:
 - Point particles: no dimensions; just translational motion retained
 - Rigid bodies: translation and rotation motion. No deformations allowed
- Of course, other types of mechanical models exist (e.g. deformable bodies)

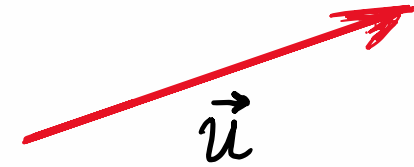
Physical quantities

- Anything that we can measure or describe about a mechanical system is a physical quantity
 - Scalar quantities are just a magnitude (and its units). Examples: mechanical energy, mass, time
 - Vector quantities not only have magnitude and units, but also direction. Their component representation in a vector basis is a column of numbers. Examples: position, velocity, acceleration; angular velocity; force
 - Vectors are sometimes classified into free, sliding, and fixed vectors, depending whether their meaning change if we move their origin or not
 - Tensor quantities represent linear relations between vectors, scalars, and other tensors. They also have units. Their representation in a vector basis is a matrix of numbers. Examples: the tensor of inertia of a rigid body.

Physical quantities do not depend on the observer,
nor his/her frame of reference or velocity

Vector spaces

- Many of the objects of interest in Mechanics are vectors from *vector spaces* over $\mathbb{R}$ under vector addition and multiplication by a scalar. We are particularly interested in $\mathbb{R}^2$ and $\mathbb{R}^3$.
- Important operations with vectors are defined:
 - Dot product (or scalar product)
 - Cross product (or vector product)



A reminder of vector spaces, vector operations, and vector properties is included in the lecture notes. Read and verify that you deeply master all this material!

- Important aspects of vectors:
 - We say a vector is *unitary* if its magnitude is equal to 1
 - We can write any vector $\mathbf{b}$ as the product of a scalar and a unit vector. The scalar encodes its *magnitude*, while the unit vector encodes its *direction*:
$$\mathbf{b} = b\mathbf{e}_b$$
 - If two vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal (a.k.a. perpendicular or normal), their dot product is zero: $\mathbf{a} \cdot \mathbf{b} = 0$
 - If two vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel, their cross product is zero: $\mathbf{a} \times \mathbf{b} = \mathbf{0}$

Vector bases and vector components

- To be able to operate with vectors, we first select a set of N linearly independent vectors from the vector space, where N is its dimension. This is a *vector basis*, B
- In $\mathbb{R}^3$ we choose three vectors $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$ (in this order) such that
 - They are unitary (magnitude equal to 1)
 - They are mutually orthogonal (i.e. perpendicular)
 - $\mathbf{i}_0 \times \mathbf{j}_0 = \mathbf{k}_0$
- Vector bases that satisfy these three conditions are called *orthonormal* (orthogonal and magnitude-normalized to 1), and *right-handed*
- Any vector can be expressed in terms of its *vector components* in this basis:

$$\mathbf{b} = b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0$$

- Multiple vector bases can be defined. To avoid ambiguity, indices are used: $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, $B_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, etc.
 - Always define the relations among the basis vectors that you use

The meaning of a vector or tensor does not depend on the vector basis used to represent it.

Vector bases and vector components

- If we have two bases $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, $B_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ we may write:

$$\begin{cases} \mathbf{i}_1 = \ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0, \\ \mathbf{j}_1 = \ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0, \\ \mathbf{k}_1 = \ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0. \end{cases}$$

Change-of-basis relations

- Any vector $\mathbf{b}$ may be expressed in B_0 or B_1 :

$$\begin{aligned} \mathbf{b} = b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0 &= b_{x1}\mathbf{i}_1 + b_{y1}\mathbf{j}_1 + b_{z1}\mathbf{k}_1 = b_{x1}(\ell_{x1,x0}\mathbf{i}_0 + \ell_{x1,y0}\mathbf{j}_0 + \ell_{x1,z0}\mathbf{k}_0) \\ &+ b_{y1}(\ell_{y1,x0}\mathbf{i}_0 + \ell_{y1,y0}\mathbf{j}_0 + \ell_{y1,z0}\mathbf{k}_0) \\ &+ b_{z1}(\ell_{z1,x0}\mathbf{i}_0 + \ell_{z1,y0}\mathbf{j}_0 + \ell_{z1,z0}\mathbf{k}_0). \end{aligned}$$

- Therefore:

$$b_{x0} = \ell_{x1,x0}b_{x1} + \ell_{y1,x0}b_{y1} + \ell_{z1,x0}b_{z1},$$

$$b_{y0} = \ell_{x1,y0}b_{x1} + \ell_{y1,y0}b_{y1} + \ell_{z1,y0}b_{z1},$$

$$b_{z0} = \ell_{x1,z0}b_{x1} + \ell_{y1,z0}b_{y1} + \ell_{z1,z0}b_{z1}.$$

Change-of-basis matrix

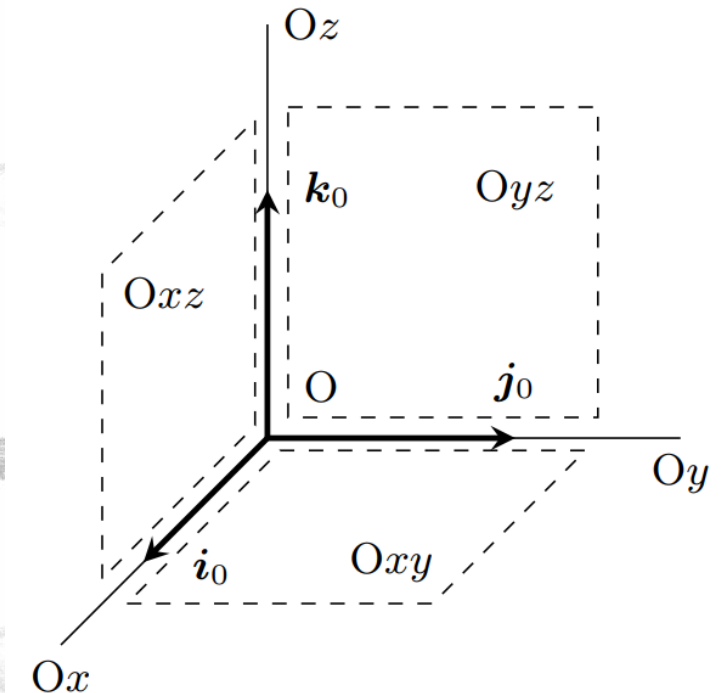
$[{}_0R_1]$

- In matrix form:

$$\begin{bmatrix} b_{x0} \\ b_{y0} \\ b_{z0} \end{bmatrix}_0 = \begin{bmatrix} \ell_{x1,x0} & \ell_{y1,x0} & \ell_{z1,x0} \\ \ell_{x1,y0} & \ell_{y1,y0} & \ell_{z1,y0} \\ \ell_{x1,z0} & \ell_{y1,z0} & \ell_{z1,z0} \end{bmatrix} \cdot \begin{bmatrix} b_{x1} \\ b_{y1} \\ b_{z1} \end{bmatrix}_1$$

Reference frames

- The motion of an object can only be described relative to something else. We must clearly define what is our *reference frame* S before we begin our analysis.
 - As before, we may define more than one reference frame. Subindices are used to eliminate ambiguity
- A reference frame $S_0 = \{O; B_0\}$ is composed of
 - An origin point, O . This is the point from where the observer measures the position of all other points of space
 - A vector basis, B_0 . This allows defining the orientation or attitude of the observer in space, as well as its rotation
- A reference frame defines naturally the coordinated axes Ox, Oy, Oz and the coordinated planes Oxy, Oxz, Oyz
 - Sometimes the notation $Oxyz$ is used as a synonym of the reference frame S_0



Time derivative of a vector

- A vector $\mathbf{b}$ changes when its magnitude, its direction, or both change. The differential of the vector $d\mathbf{b}$ is itself a vector

$$\mathbf{b} = b\mathbf{e}_b$$

$$d\mathbf{b} = (db)\mathbf{e}_b + b(d\mathbf{e}_b)$$

- Dot-multiplying by $\mathbf{e}_b$:

$$\mathbf{e}_b \cdot (d\mathbf{b}) = db + b\mathbf{e}_b \cdot (d\mathbf{e}_b)$$

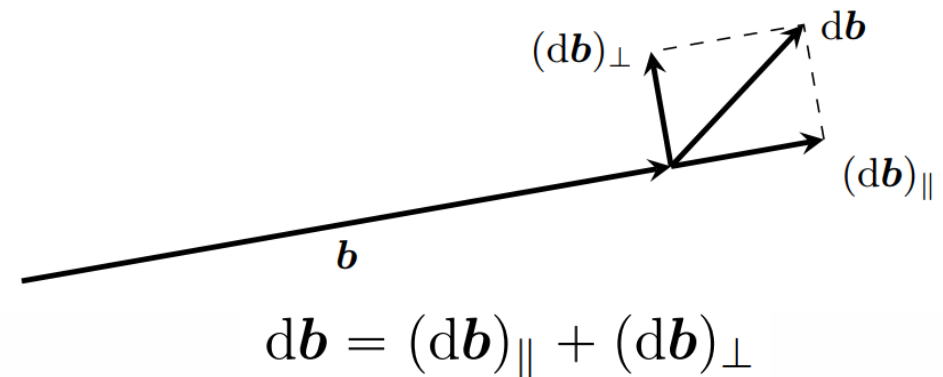
but:

$$\mathbf{e}_b \cdot (d\mathbf{e}_b) = \frac{1}{2}d(\mathbf{e}_b \cdot \mathbf{e}_b) = \frac{1}{2}d(e_b^2) = \frac{1}{2}d1 = 0$$

- We identify:

$$(d\mathbf{b})_{\parallel} = db\mathbf{e}_b \quad (d\mathbf{b})_{\perp} = b(d\mathbf{e}_b)$$

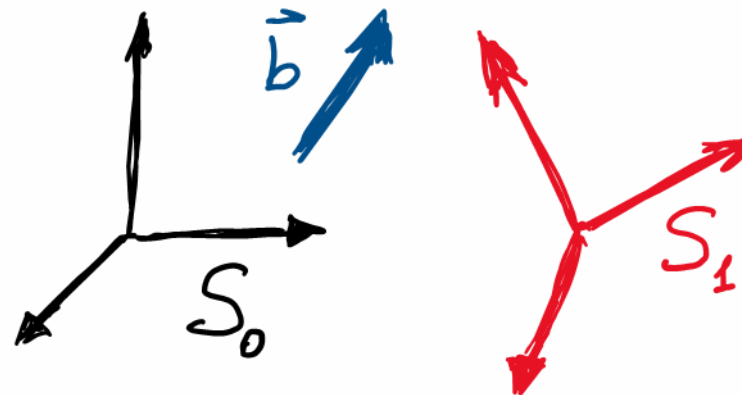
- The component of $d\mathbf{b}$ along $\mathbf{e}_b$ describes changes in the magnitude of $\mathbf{b}$
- The component of $d\mathbf{b}$ perpendicular to $\mathbf{e}_b$ describes changes in the direction of $\mathbf{b}$
- Corollary: the derivative of a unit vector like $\mathbf{e}_b$ (magnitude is constant and equal to 1) is always $\perp$ to it



Time derivative of a vector

- The differential, and therefore the change rate of a vector depends on the reference frame it is observed from. To avoid any ambiguity, we will identify the reference frame in which the time derivative is taken with a subindex:

$$\left. \frac{d\mathbf{b}}{dt} \right|_0$$



In general:

$$\left. \frac{d\mathbf{b}}{dt} \right|_0 \neq \left. \frac{d\mathbf{b}}{dt} \right|_1$$

- The vectors of basis B_0 do not vary in S_0 :

$$\left. \frac{d\mathbf{i}_0}{dt} \right|_0 = \mathbf{0}$$

$$\left. \frac{d\mathbf{j}_0}{dt} \right|_0 = \mathbf{0}$$

$$\left. \frac{d\mathbf{k}_0}{dt} \right|_0 = \mathbf{0}$$

$$\mathbf{b} = b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0 = b_{x1}\mathbf{i}_1 + b_{y1}\mathbf{j}_1 + b_{z1}\mathbf{k}_1$$

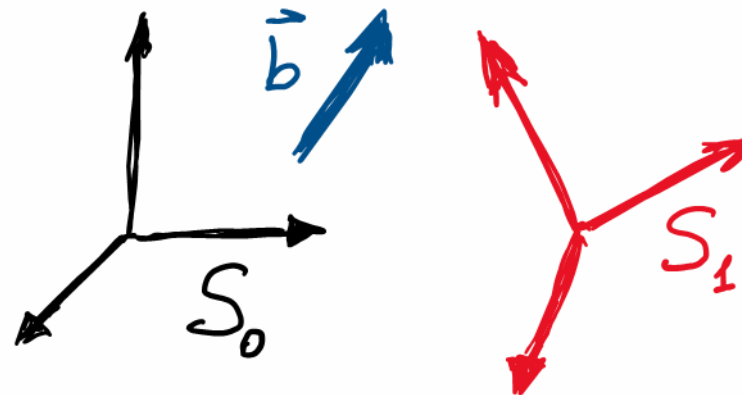
- If we know the components of $\mathbf{b}$ in B_0 , we may compute its time derivative in S_0 as:

$$\left. \frac{d\mathbf{b}}{dt} \right|_0 = \frac{db_{x0}}{dt}\mathbf{i}_0 + \frac{db_{y0}}{dt}\mathbf{j}_0 + \frac{db_{z0}}{dt}\mathbf{k}_0$$

Time derivative of a vector

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$$\left. \frac{d\mathbf{k}_0}{dt} \right|_0 = \mathbf{0}$$

$$\mathbf{b} = b_{x0}\mathbf{i}_0 + b_{y0}\mathbf{j}_0 + b_{z0}\mathbf{k}_0 = \underline{b_{x1}\mathbf{i}_1 + b_{y1}\mathbf{j}_1 + b_{z1}\mathbf{k}_1}$$

- If we know the components of $\mathbf{b}$ in B_1 , we may compute its time derivative in S_0 as:

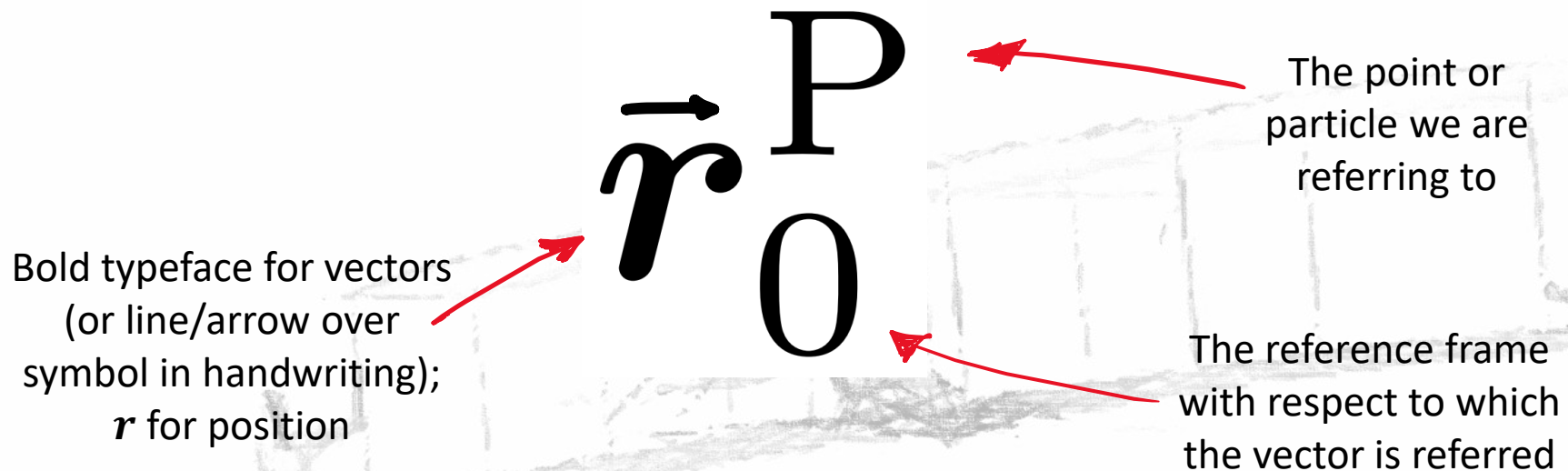
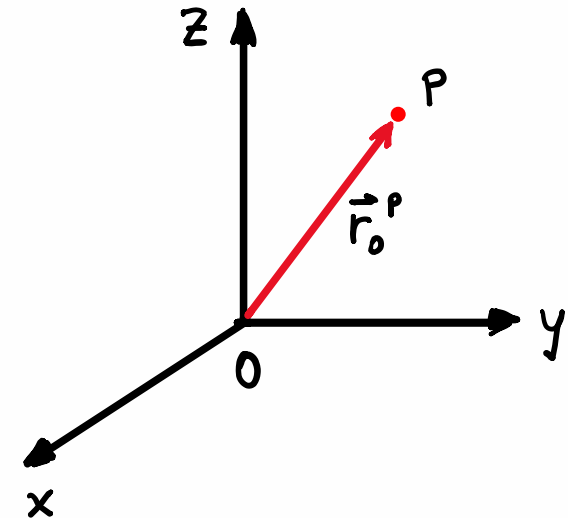
$$\left. \frac{d\mathbf{b}}{dt} \right|_0 = \frac{db_{x1}}{dt}\mathbf{i}_1 + \frac{db_{y1}}{dt}\mathbf{j}_1 + \frac{db_{z1}}{dt}\mathbf{k}_1 + \underline{b_{x1} \left. \frac{d\mathbf{i}_1}{dt} \right|_0 + b_{y1} \left. \frac{d\mathbf{j}_1}{dt} \right|_0 + b_{z1} \left. \frac{d\mathbf{k}_1}{dt} \right|_0}$$

Position vector

- We define the position vector of a point P with respect to a reference frame S_0 as:

$$\vec{r}_0^P = \mathbf{OP}$$

- Important: watch the nomenclature:



The position vectors with respect to S_0 and S_1 are different if their origins are different

Velocity and acceleration vectors

- The position vector may change in time as seen from S_0 . We define the velocity vector of a point P with respect to a reference frame S_0 as:

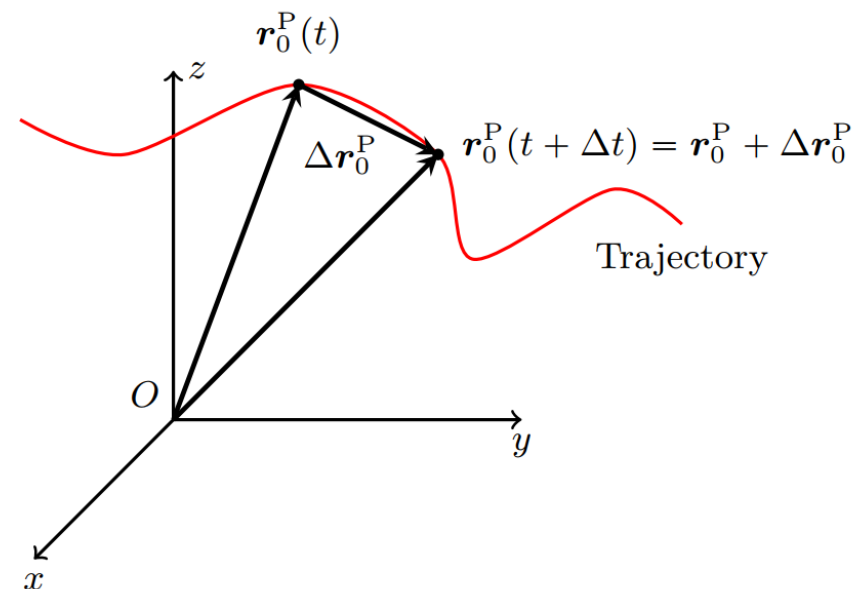
$$\mathbf{v}_0^P = \lim_{\Delta t \rightarrow 0} \left. \frac{(\mathbf{r}_0^P + \Delta \mathbf{r}_0^P) - \mathbf{r}_0^P}{\Delta t} \right|_0 = \lim_{\Delta t \rightarrow 0} \left. \frac{\Delta \mathbf{r}_0^P}{\Delta t} \right|_0 = \left. \frac{d\mathbf{r}_0^P}{dt} \right|_0$$

- Its magnitude (scalar) is sometimes called speed or celerity
- Likewise, the velocity vector may change in time. The acceleration vector of a point P with respect to a reference frame S_0 as is:

$$\mathbf{a}_0^P = \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 = \left. \frac{d^2\mathbf{r}_0^P}{dt^2} \right|_0$$

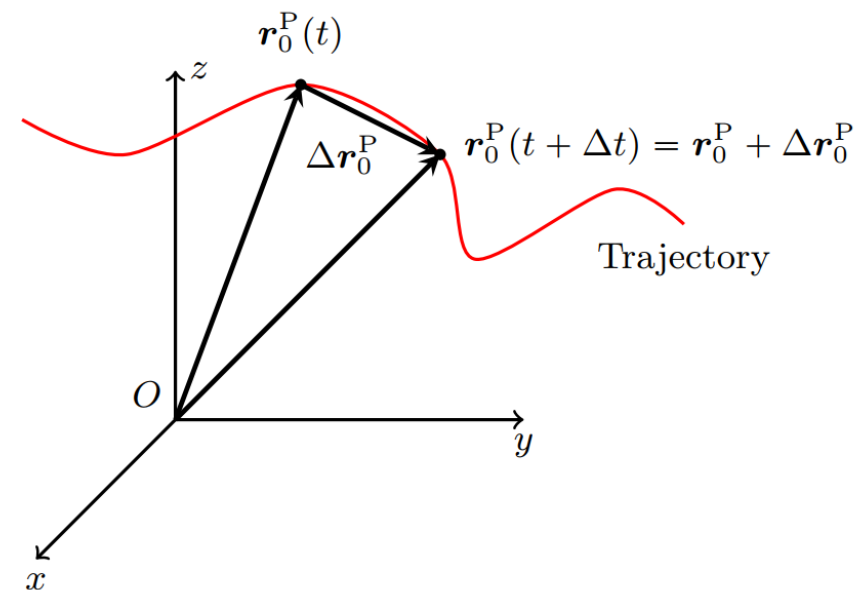
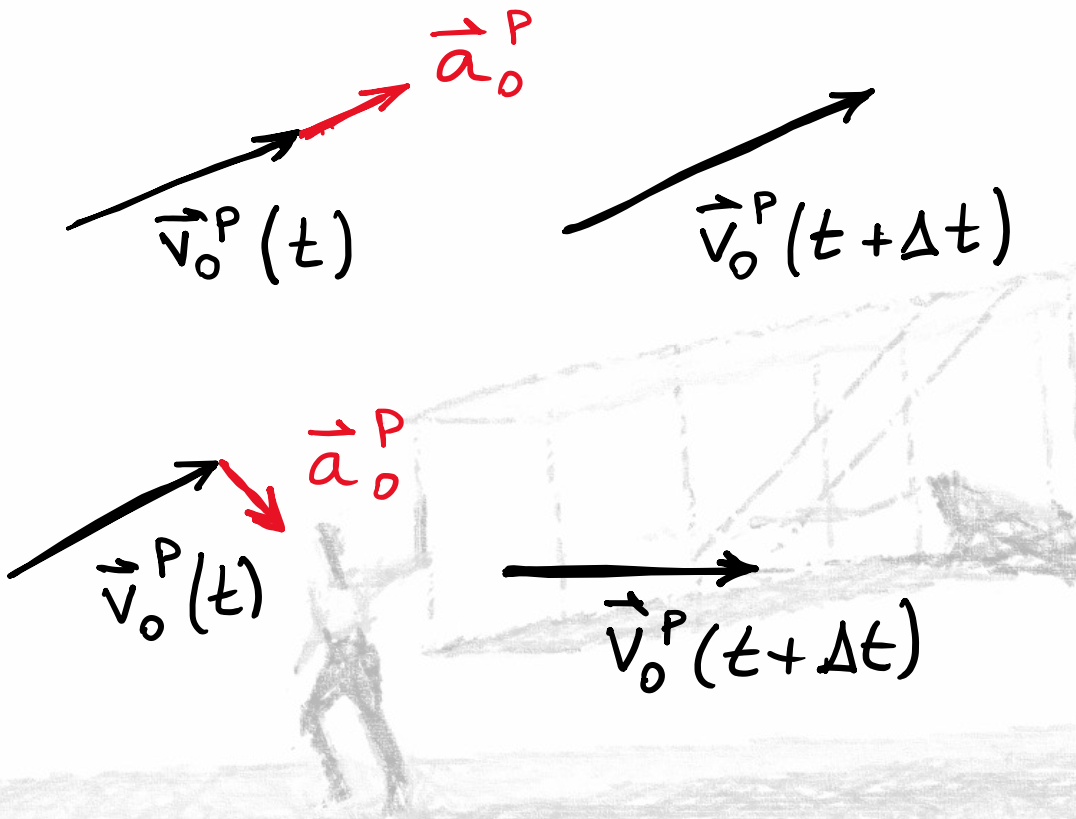
The velocity and acceleration vectors with respect to S_0 and S_1 are different if their origins are translating and/or rotating with respect to each other

Beware of weird hybrids like $\left. \frac{d\mathbf{r}_0^P}{dt} \right|_1$!



Velocity and acceleration vectors

- The velocity vector points always in the direction of the differential $d\mathbf{r}_0^P$, which is tangent to the trajectory described by the point
- Over time, the velocity vector $\mathbf{v}_0^P$ may change in...
 - Magnitude: this is associated with an acceleration vector parallel to $\mathbf{v}_0^P$
 - Direction: this is associated with an acceleration vector perpendicular to $\mathbf{v}_0^P$



Coordinate systems

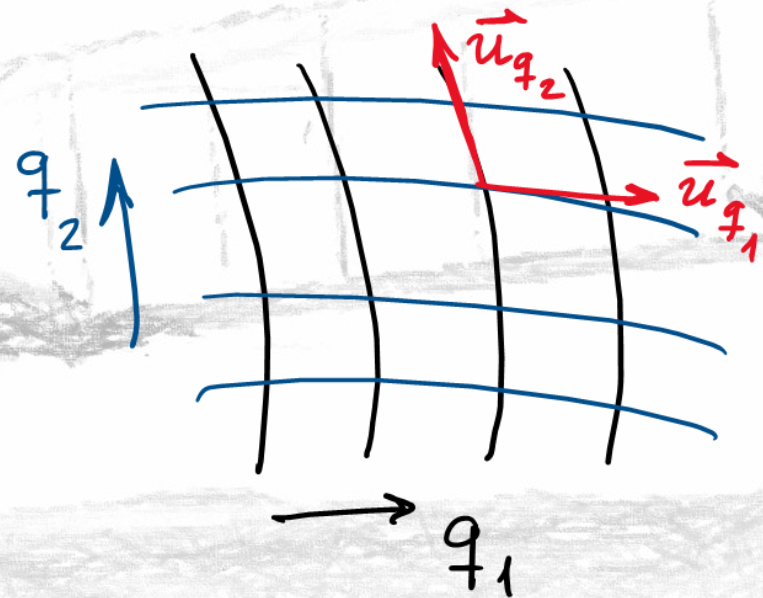
- The position of a point in space ($\mathbb{R}^3$) may be given in terms of three scalar quantities, or *coordinates*. These make up a coordinate system

$$C = \{q_1, q_2, q_3\}$$

- Examples of classical coordinate systems are *Cartesian*, *cylindrical*, and *spherical*
- In each coordinate system we may define an associated vector basis by differentiating the position vector with respect to each coordinate:

$$\mathbf{r}_0^P = \mathbf{r}_0^P(q_1, q_2, q_3) \quad \rightarrow \quad \mathbf{u}_{q_1} = \frac{\partial \mathbf{r}_0^P}{\partial q_1}; \quad \mathbf{u}_{q_2} = \frac{\partial \mathbf{r}_0^P}{\partial q_2}; \quad \mathbf{u}_{q_3} = \frac{\partial \mathbf{r}_0^P}{\partial q_3}$$

- These vectors may be non-unitary, but they can always be normalized if desired
- If $\mathbf{u}_{q_1}$, $\mathbf{u}_{q_2}$, and $\mathbf{u}_{q_3}$ are mutually orthogonal, we say that C is an *orthogonal coordinate system*. Cartesian, cylindrical, and spherical coordinates are orthogonal.



Cartesian coordinates

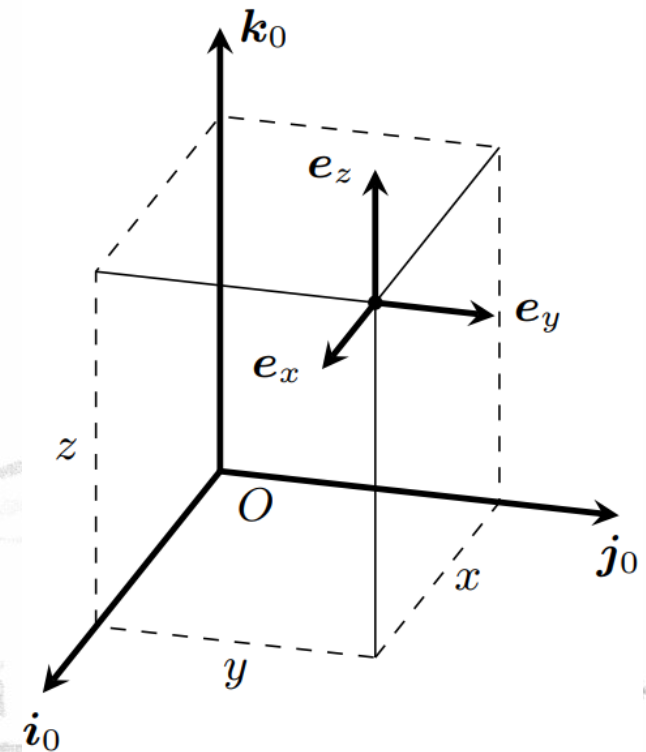
- Coordinates $\{x, y, z\}$; defined “by default” in each reference frame.
- The associated vector basis is just the set of vectors $\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0$
- Kinematics in Cartesian coordinates:

$$\mathbf{r}_0^P = x\mathbf{i}_0 + y\mathbf{j}_0 + z\mathbf{k}_0$$

Cartesian coordinates are the only ones where the coordinates equal the components of the position vector (in the associated basis)

$$\mathbf{v}_0^P = \left. \frac{d\mathbf{r}_0^P}{dt} \right|_0 = \dot{x}\mathbf{i}_0 + \dot{y}\mathbf{j}_0 + \dot{z}\mathbf{k}_0$$

$$\mathbf{a}_0^P = \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 = \ddot{x}\mathbf{i}_0 + \ddot{y}\mathbf{j}_0 + \ddot{z}\mathbf{k}_0$$



Cylindrical coordinates

- Instead of x, y we define R, θ . Coordinate z is still the same

$$\begin{cases} x = R \cos \theta \\ y = R \sin \theta \end{cases} \quad \begin{cases} R = \sqrt{x^2 + y^2} \\ \theta = \arctan(y/x) \end{cases}$$

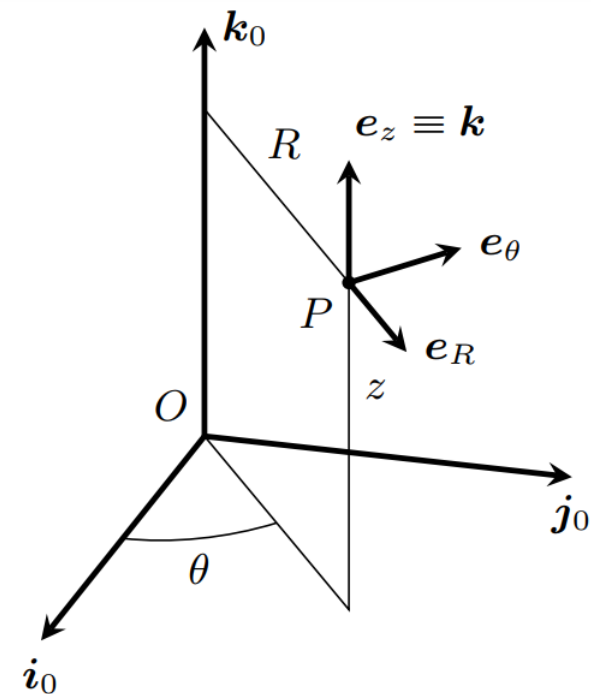
- The associated vector basis B_C is:

$$\begin{cases} e_R = \cos \theta \mathbf{i}_0 + \sin \theta \mathbf{j}_0, \\ e_\theta = -\sin \theta \mathbf{i}_0 + \cos \theta \mathbf{j}_0, \\ e_z = \mathbf{k}_0, \end{cases}$$

- These vectors are not the same at all points of space: they *vary* with coordinate θ . The derivatives are:

$$\begin{aligned} \frac{\partial e_R}{\partial \theta} &= -\sin \theta \mathbf{i}_0 + \cos \theta \mathbf{j}_0 = e_\theta, \\ \frac{\partial e_\theta}{\partial \theta} &= -\cos \theta \mathbf{i}_0 - \sin \theta \mathbf{j}_0 = -e_R. \end{aligned}$$

The vectors of the associated basis of a coordinate system need not be the same at all points of space



Cylindrical coordinates

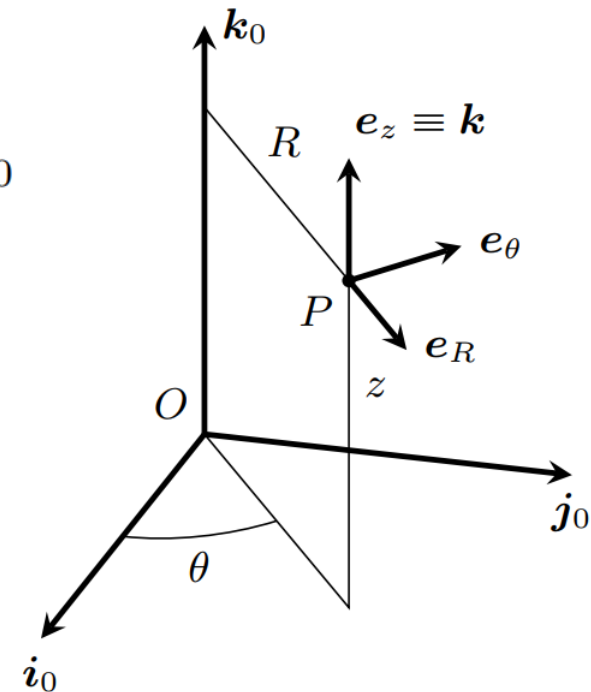
➤ Kinematics in cylindrical coordinates:

$$\mathbf{r}_0^P = R\mathbf{e}_R + z\mathbf{e}_z = R\cos\theta\mathbf{i}_0 + R\sin\theta\mathbf{j}_0 + z\mathbf{k}_0,$$

$$\begin{aligned}\mathbf{v}_0^P &= \dot{R}\mathbf{e}_R + R\dot{\theta}\mathbf{e}_\theta + \dot{z}\mathbf{e}_z \\ &= (\dot{R}\cos\theta - R\dot{\theta}\sin\theta)\mathbf{i}_0 + (\dot{R}\sin\theta + R\dot{\theta}\cos\theta)\mathbf{j}_0 + \dot{z}\mathbf{k}_0,\end{aligned}$$

$$\begin{aligned}\mathbf{a}_0^P &= (\ddot{R} - R\dot{\theta}^2)\mathbf{e}_R + (R\ddot{\theta} + 2\dot{R}\dot{\theta})\mathbf{e}_\theta + \ddot{z}\mathbf{e}_z \\ &= (\ddot{R}\cos\theta - R\dot{\theta}^2\cos\theta - 2\dot{R}\dot{\theta}\sin\theta - R\ddot{\theta}\sin\theta)\mathbf{i}_0 \\ &\quad + (\ddot{R}\sin\theta - R\dot{\theta}^2\sin\theta + 2\dot{R}\dot{\theta}\cos\theta + R\ddot{\theta}\cos\theta)\mathbf{j}_0 \\ &\quad + \ddot{z}\mathbf{k}_0.\end{aligned}$$

Observe that we can write each vector using R, θ, z , and we can also choose to use the $B_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ vector basis or the $B_C = \{\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z\}$ vector basis



Spherical coordinates

➤ Spherical coordinates r, θ, ϕ :

$$\begin{cases} x = r \cos \phi \cos \theta \\ y = r \cos \phi \sin \theta \\ z = r \sin \phi \end{cases} \quad \begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \theta = \arctan(y/x) \\ \phi = \arctan(z/\sqrt{x^2 + y^2}) \end{cases}$$

➤ Associated vector basis B_S :

$$\begin{cases} \mathbf{e}_r = \cos \phi \cos \theta \mathbf{i}_0 + \cos \phi \sin \theta \mathbf{j}_0 + \sin \phi \mathbf{k}_0, \\ \mathbf{e}_\theta = -\sin \theta \mathbf{i}_0 + \cos \theta \mathbf{j}_0, \\ \mathbf{e}_\phi = -\sin \phi \cos \theta \mathbf{i}_0 - \sin \phi \sin \theta \mathbf{j}_0 + \cos \phi \mathbf{k}_0, \end{cases}$$

▪ Derivatives:

Note: sometimes spherical coordinates are defined with $\varphi = \pi/2 - \phi$ instead of ϕ . Always check!

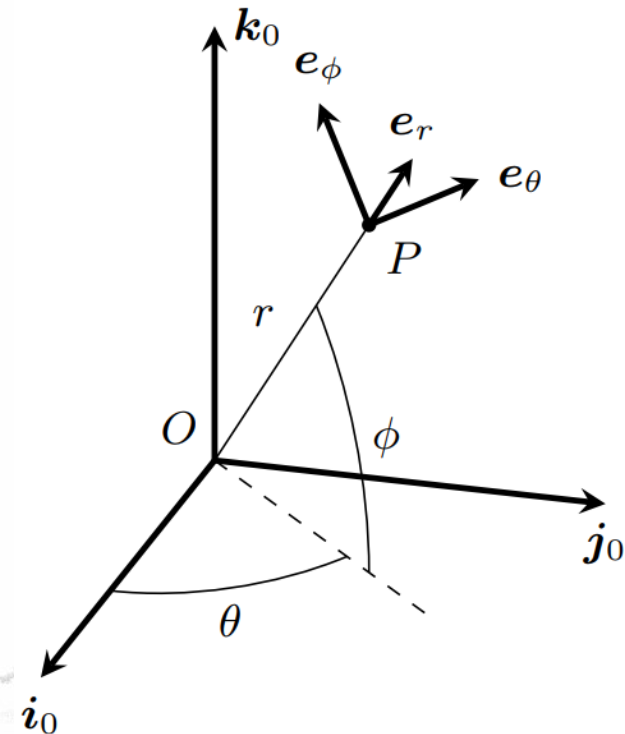
$$\frac{\partial \mathbf{e}_r}{\partial \theta} = \cos \phi (-\sin \theta \mathbf{i}_0 + \cos \theta \mathbf{j}_0) = \cos \phi \mathbf{e}_\theta,$$

$$\frac{\partial \mathbf{e}_\theta}{\partial \theta} = -\cos \theta \mathbf{i}_0 - \sin \theta \mathbf{j}_0 = -\cos \phi \mathbf{e}_r + \sin \phi \mathbf{e}_\phi,$$

$$\frac{\partial \mathbf{e}_\phi}{\partial \theta} = \sin \phi (\sin \theta \mathbf{i}_0 - \cos \theta \mathbf{j}_0) = -\sin \phi \mathbf{e}_\theta.$$

$$\frac{\partial \mathbf{e}_r}{\partial \phi} = -\sin \phi \cos \theta \mathbf{i}_0 - \sin \phi \sin \theta \mathbf{j}_0 + \cos \phi \mathbf{k}_0 = \mathbf{e}_\phi,$$

$$\frac{\partial \mathbf{e}_\phi}{\partial \phi} = -\cos \phi \cos \theta \mathbf{i}_0 - \cos \phi \sin \theta \mathbf{j}_0 - \sin \phi \mathbf{k}_0 = -\mathbf{e}_r.$$



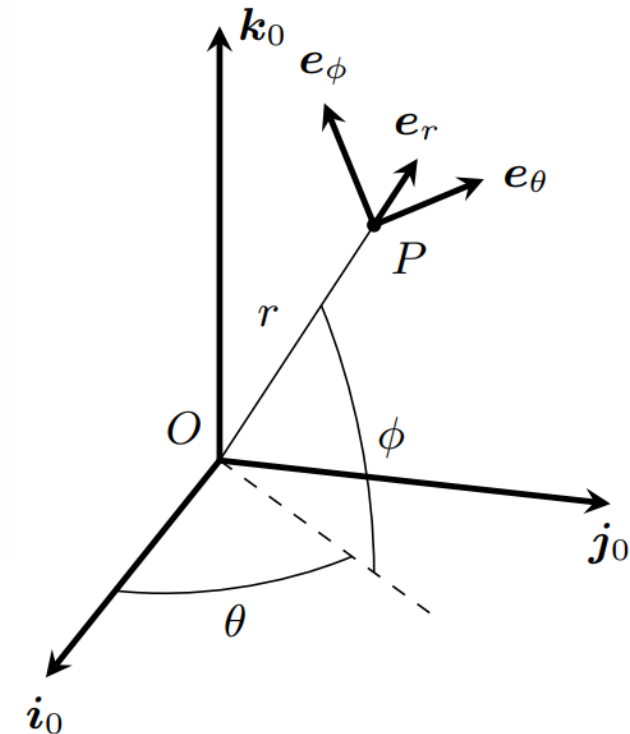
Spherical coordinates

➤ Kinematics in spherical coordinates

$$\mathbf{r}_0^P = r\mathbf{e}_r = r \cos \phi \cos \theta \mathbf{i}_0 + r \cos \phi \sin \theta \mathbf{j}_0 + r \sin \phi \mathbf{k}_0,$$

$$\begin{aligned} \mathbf{v}_0^P &= \dot{r}\mathbf{e}_r + r\dot{\theta} \cos \phi \mathbf{e}_\theta + r\dot{\phi} \mathbf{e}_\phi \\ &= (\dot{r} \cos \phi \cos \theta - r\dot{\phi} \sin \phi \cos \theta - r\dot{\theta} \cos \phi \sin \theta) \mathbf{i}_0 \\ &\quad + (\dot{r} \cos \phi \sin \theta - r\dot{\phi} \sin \phi \sin \theta + r\dot{\theta} \cos \phi \cos \theta) \mathbf{j}_0 \\ &\quad + (\dot{r} \sin \phi + r\dot{\phi} \cos \phi) \mathbf{k}_0 \end{aligned}$$

$$\begin{aligned} \mathbf{a}_0^P &= (\ddot{r} - r\dot{\theta}^2 \cos^2 \phi - r\dot{\phi}^2) \mathbf{e}_r \\ &\quad + (2\dot{r}\dot{\theta} \cos \phi + r\ddot{\theta} \cos \phi - 2r\dot{\theta}\dot{\phi} \sin \phi) \mathbf{e}_\theta \\ &\quad + (2\dot{r}\dot{\phi} + r\ddot{\phi} + r\dot{\theta}^2 \cos \phi \sin \phi) \mathbf{e}_\phi \\ &= (\ddot{r} \cos \phi \cos \theta - 2\dot{r}\dot{\phi} \sin \phi \cos \theta - 2\dot{r}\dot{\theta} \cos \phi \sin \theta - r\ddot{\phi} \sin \phi \cos \theta \\ &\quad - r\dot{\phi}^2 \cos \phi \cos \theta + 2r\dot{\phi}\dot{\theta} \sin \phi \sin \theta - r\ddot{\theta} \cos \phi \sin \theta - r\dot{\theta}^2 \cos \phi \cos \theta) \mathbf{i}_0 \\ &\quad + (\ddot{r} \cos \phi \sin \theta - 2\dot{r}\dot{\phi} \sin \phi \sin \theta + 2\dot{r}\dot{\theta} \cos \phi \cos \theta - r\ddot{\phi} \sin \phi \sin \theta \\ &\quad - r\dot{\phi}^2 \cos \phi \sin \theta - 2r\dot{\phi}\dot{\theta} \sin \phi \cos \theta + r\ddot{\theta} \cos \phi \cos \theta - r\dot{\theta}^2 \cos \phi \sin \theta) \mathbf{j}_0 \\ &\quad + (\ddot{r} \sin \phi + 2\dot{r}\dot{\phi} \cos \phi + r\ddot{\phi} \cos \phi - r\dot{\phi}^2 \sin \phi) \mathbf{k}_0. \end{aligned}$$



Generalized coordinates and degrees of freedom

- We do not need to restrict ourselves to Cartesian, cylindrical or spherical coordinates: any three parameters that unambiguously locate a point in space will do as a coordinate system: angles, areas, distances...
 - In many situations, a point particle cannot move freely in $\mathbb{R}^3$, but it is constrained to stay on a surface (two-dimensional object) or a line (one-dimensional object).
 - In other situations, there are several point particles; their motion may be related or constrained somehow, e.g. particles connected by a string.
- In all these situations, we are interested in describing the possible *configurations* of the system using a minimum number of coordinates.
 - Particle on a surface: $\mathbf{r}_0^P = \mathbf{r}_0^P(q_1, q_2)$
 - Particle on a curve: $\mathbf{r}_0^P = \mathbf{r}_0^P(q_1)$ } Same number of coordinates as the dimension of the underlying object
- In these cases we speak of *generalized coordinates*

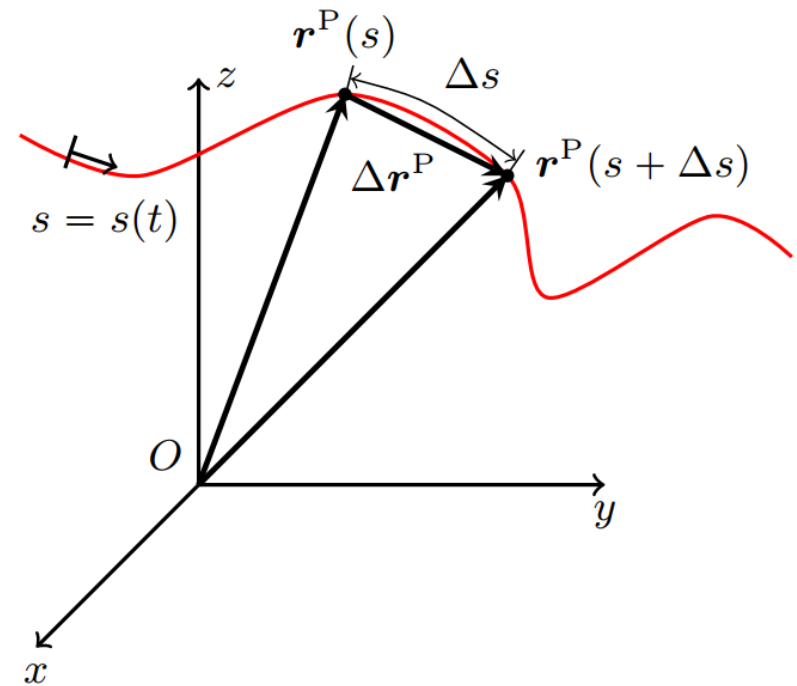
The minimum number of coordinates needed is the *number of configuration degrees of freedom of the system*, and the space of all possible values of the selected coordinates is the *configuration space*

Trajectory

- The trajectory or path is the line described by a point P as it moves, as seen from a particular reference frame
- We define an *arc length coordinate* “ s ” that measures the distance traveled along the trajectory by P from a chosen origin
 - If we know the shape of the trajectory, we may write the position of P as a function of s :
$$\mathbf{r}_0^P = \mathbf{r}_0^P(s)$$
 - Then, if we determine $s = s(t)$, we have full knowledge of the motion of P
- We call *path variables* those quantities (scalars, vectors...) that only depend on the shape of the trajectory, and not on how quickly the particle P moves along it:

$$\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$$

$$\kappa \quad \rho \quad \tau$$



Intrinsic vector basis

- Consider the increment

$$\Delta \mathbf{r}^P = \mathbf{r}^P(s + \Delta s) - \mathbf{r}^P(s)$$

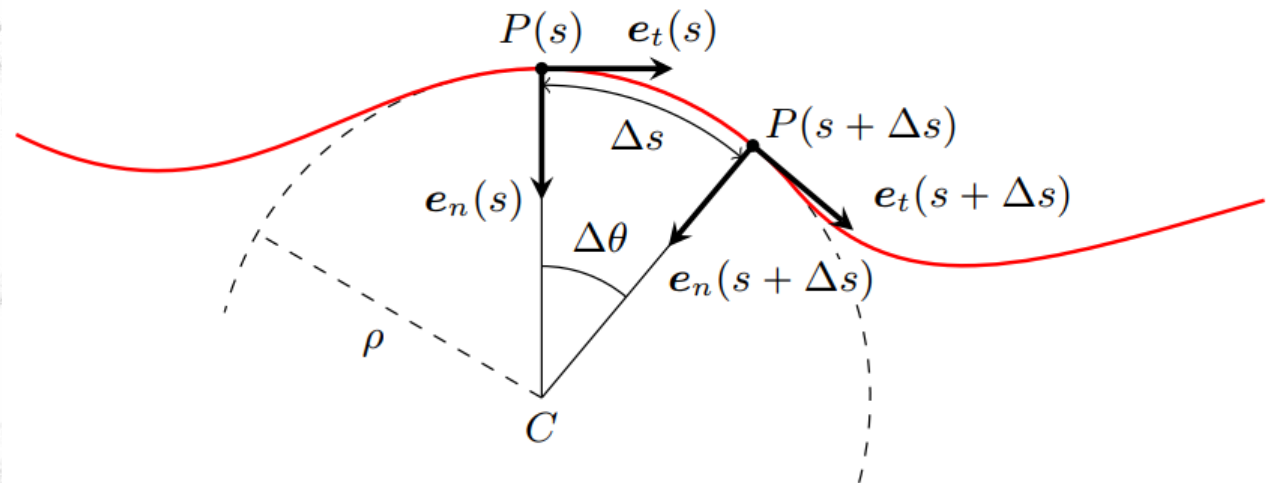
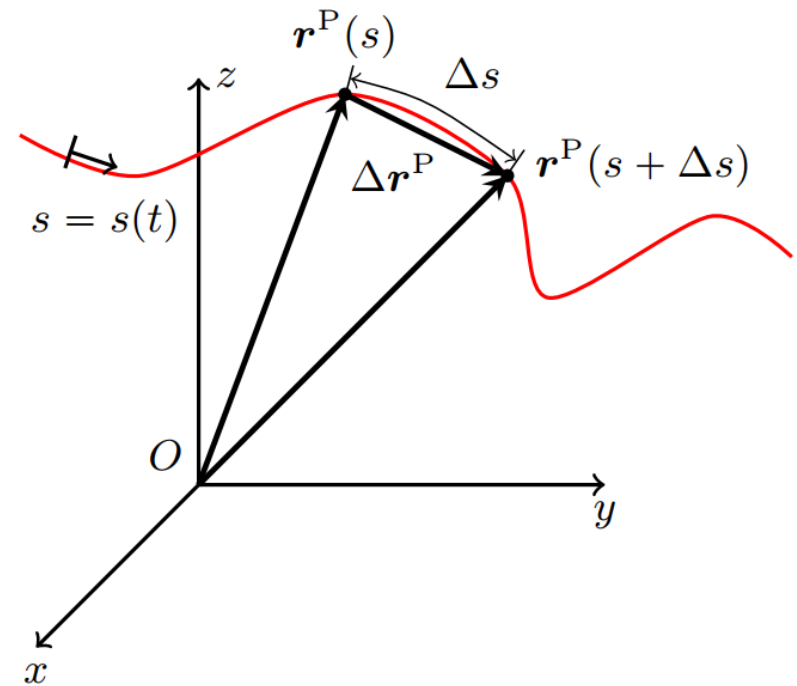
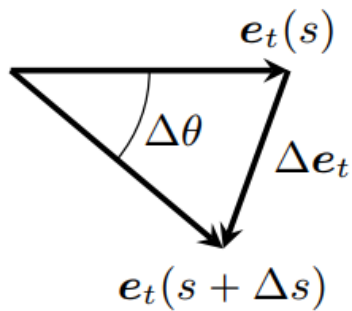
As $\Delta s \rightarrow 0$, $\Delta \mathbf{r}^P$ becomes tangent to the trajectory, and its magnitude becomes Δs

- We define the *tangent unit vector* $\mathbf{e}_t$ as:

$$\frac{d\mathbf{r}^P}{ds} = \lim_{\Delta s \rightarrow 0} \frac{\Delta \mathbf{r}^P}{\Delta s} \equiv \mathbf{e}_t(s)$$

- $\mathbf{e}_t$ is always tangent to the trajectory

- How does $\mathbf{e}_t$ change with s ? In the limit $\Delta s \rightarrow 0$, $\Delta \mathbf{e}_t$ becomes *normal* to $\mathbf{e}_t$



Intrinsic vector basis

- We define the *normal unit vector* $\mathbf{e}_n$ as:

$$\mathbf{e}_n(s) = \frac{d\mathbf{e}_t}{ds} \bigg/ \left| \frac{d\mathbf{e}_t}{ds} \right|$$

$\mathbf{e}_n$ is not well defined in straight segments of the trajectory, because the derivative of $\mathbf{e}_t$ is zero

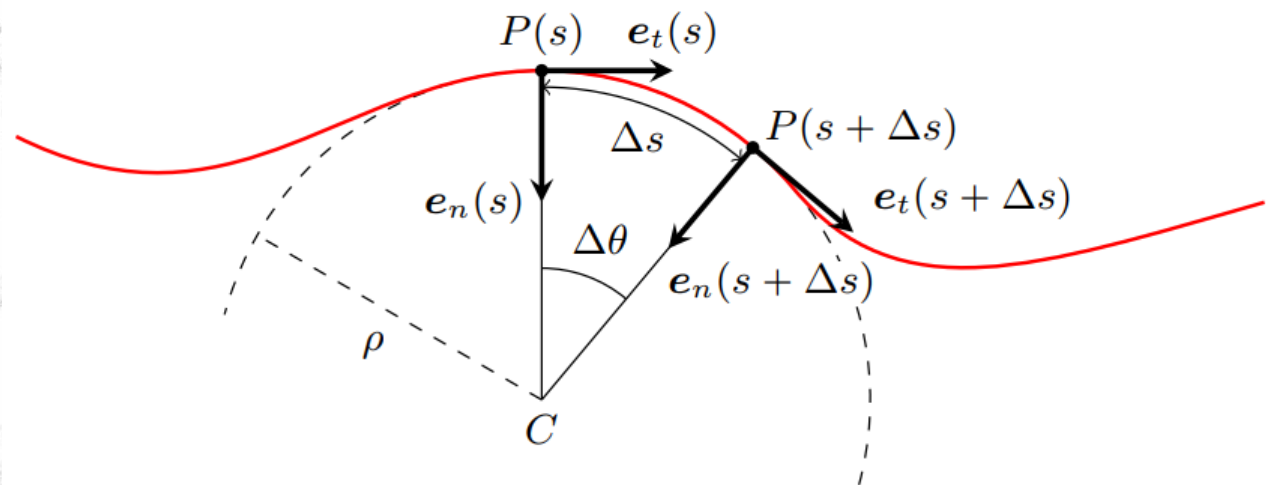
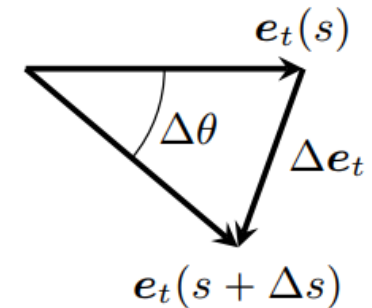
- We divide by the magnitude of the derivative to obtain a unit vector
- The magnitude of the derivative is the *curvature* κ ; its inverse is the radius of curvature ρ :

$$\left| \frac{d\mathbf{e}_t}{ds} \right| = \frac{d\theta}{ds} = \frac{1}{\rho} = \kappa$$

- The vectors $\mathbf{e}_t$ and $\mathbf{e}_n$ define the *osculating plane*

- Locally, the trajectory can be approximated by a segment of an *osculating circle* of radius ρ and center C

- $\mathbf{e}_n$ always points toward C



Intrinsic vector basis

- To complete an orthonormal, right-handed vector basis we define the *binormal unit vector* as

$$\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$$

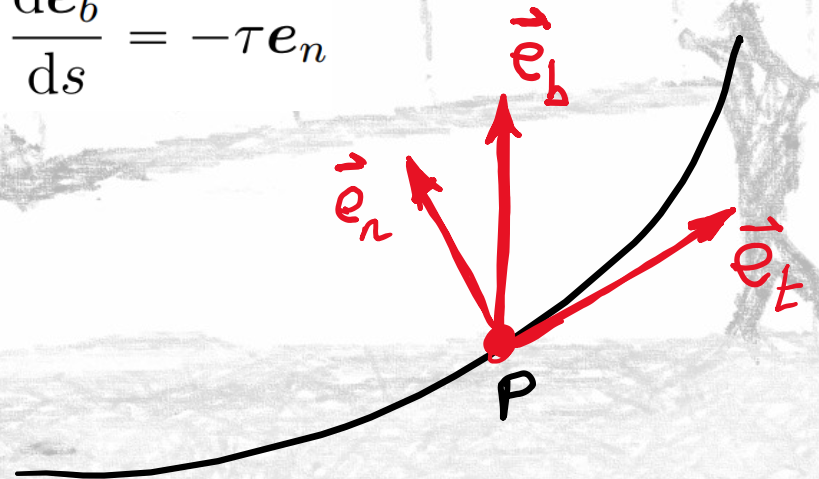
- The basis $B_P = \{\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b\}$ is the *intrinsic vector basis* at point P; $S_P = \{P; B_P\}$ is the *intrinsic reference frame*

- $\mathbf{e}_t$ is always tangent to the trajectory
- $\mathbf{e}_n$ is always normal to the trajectory, pointing toward the local center of curvature
- $\mathbf{e}_b$ is perpendicular to the osculating plane, so that $\mathbf{e}_b = \mathbf{e}_t \times \mathbf{e}_n$

- The derivatives of these unit vectors with respect to s are given by the *Frenet-Serret formulas*:

$$\frac{d\mathbf{e}_t}{ds} = \kappa \mathbf{e}_n; \quad \frac{d\mathbf{e}_n}{ds} = -\kappa \mathbf{e}_t + \tau \mathbf{e}_b; \quad \frac{d\mathbf{e}_b}{ds} = -\tau \mathbf{e}_n$$

- κ is the *curvature*
- τ is the *torsion*. It is only non-zero in 3D curves



Tangential and normal acceleration

- Using s and the intrinsic vector basis we can write $\mathbf{v}^P$ as:

$$\mathbf{v}^P = \frac{d\mathbf{r}^P}{dt} = \frac{d\mathbf{r}^P}{ds} \frac{ds}{dt} = \dot{s} \mathbf{e}_t$$

- The velocity vector is always tangent to the trajectory: its unit vector is $\mathbf{e}_t$
- There are no components of $\mathbf{v}^P$ along $\mathbf{e}_n$ and $\mathbf{e}_b$
- Its magnitude (the speed) is equal to $\dot{s}$

$\mathbf{v}^P$ and $\mathbf{a}^P$
always lie in the
osculating plane

- Taking derivatives, the acceleration $\mathbf{a}^P$ is:

$$\mathbf{a}^P = \frac{d\mathbf{v}^P}{dt} = \ddot{s} \mathbf{e}_t + \dot{s} \frac{d\mathbf{e}_t}{ds} \frac{ds}{dt} = \ddot{s} \mathbf{e}_t + \frac{\dot{s}^2}{\rho} \mathbf{e}_n$$

- The acceleration vector has two components: the *tangential acceleration* a_t along $\mathbf{e}_t$ and the *normal acceleration* a_n along $\mathbf{e}_n$
 - a_t is responsible for changes in the magnitude of $\mathbf{v}^P$ (i.e., the speed). If it is zero, the speed is constant
 - a_n is responsible for changes in the direction of $\mathbf{v}^P$. If it is zero, the direction of translation of P is constant
- There is no component of $\mathbf{a}^P$ along $\mathbf{e}_b$

Exercise

1. An out-of-control rocket traces an upward helical trajectory described in cartesian coordinates for $t > 0$ by

$$x = a \cos(\omega t), \quad (1)$$

$$y = a \sin(\omega t), \quad (2)$$

$$z = v_z t. \quad (3)$$

where a , v_z and ω are constants.

- (a) Express the position, velocity and acceleration vectors in the cartesian x, y, z system using unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$.
- (b) Express the position, velocity and acceleration vectors in the cylindrical coordinate system using unit vectors $\mathbf{e}_R, \mathbf{e}_\theta, \mathbf{e}_z$.
- (c) Express the velocity and acceleration vectors using the intrinsic s, ρ coordinates and vector basis $\mathbf{e}_t, \mathbf{e}_n, \mathbf{e}_b$.
- (d) Compute the modulus of the velocity vector
- (e) Express the evolution of the arc-length coordinate as a function of time
- (f) Write down the trajectory as a function of the arc-length coordinate
- (g) Where is the center of curvature for this curve when the particle is located at $(x, y, z) = (a, 0, 0)$?

Thank you!

